ABHIRAM R S

Kannur, Kerala | +91 7736110891 | abhiramvellur40@gmail.com | LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Integrated M.Sc. student specialising in Artificial Intelligence and Machine Learning with hands-on project experience in computer vision, predictive modelling, and deep learning. Built a real-time YOLO-based object detection system for railway safety, an end-to-end ML classification pipeline for fintech data, and a real-time zero-day intrusion detection system using reconstruction-error-based anomaly detection. Proficient in Python, TensorFlow, PyTorch, and Scikit-learn. Seeking an entry-level AI/ML engineering or research role to deliver impact through applied intelligence.

TECHNICAL SKILLS

Languages	Python, SQL, C, Java, HTML, JavaScript
ML / DL	TensorFlow, PyTorch, Keras, Scikit-learn, YOLO, LSTM, Attention Models
Data Analysis	Pandas, NumPy, Matplotlib
Web / Backend	Django, FastAPI, MySQL, SQLite, Redis
Tools & Env	Git, GitHub, Jupyter Notebook, Google Colab, Docker, LaTeX

PROJECTS

Zero-Day Intrusion Detection System *Python · PyTorch · FastAPI · Redis · Docker · Network Security*
github.com/AbhiR4mRs/zeroday

- Developed a **real-time zero-day intrusion detection system** that learns normal network traffic patterns and flags anomalous behavior using reconstruction error.
- Built an **attention-based LSTM autoencoder** to process sequential flow windows and detect suspicious traffic with threshold-based anomaly scoring.
- Integrated the pipeline with **Redis, FastAPI, and Docker** for live traffic buffering, inference, and deployment, making the system suitable for real-time monitoring.

Foreign Object Intrusion Detection — Railway Safety *Python · YOLO · OpenCV · Computer Vision*
github.com/AbhiR4mRs/Yolo

- Designed and deployed a **YOLO-based real-time object detection pipeline** to identify foreign objects on active railway tracks, directly supporting safety-critical infrastructure monitoring.
- Engineered a complete data workflow covering frame extraction, image annotation, model training, and live inference — achieving reliable detection under varied lighting and weather conditions.
- Architected the system for potential edge deployment, making it applicable to IoT-based railway safety networks with low-latency requirements.

Credit Card Approval Prediction *Python · Scikit-learn · Pandas · Imbalanced-Learn*
Project Link

- Built a machine learning model to predict credit card approval status using applicant demographic and financial data from a merged dataset of 1,548 records.
- Performed data cleaning and feature engineering by handling missing values, deriving **Age** and **EmploymentYears**, and removing outliers using the IQR method.
- Applied **StandardScaler**, **OneHotEncoder**, and **SMOTE** for preprocessing and class balancing, then compared multiple classifiers including Logistic Regression, Decision Tree, Random Forest, SVM, Gradient Boosting, Naive Bayes, and KNN.
- Achieved **93.43% accuracy** with **Random Forest**, and exported the trained model and preprocessing pipeline as `.pkl` files for reuse.

Hostel Management System *Python · Django · SQLite · HTML · JavaScript*
github.com/jitheshjr/Hostel_Manangement_System

- Developed a full-stack MVC web application automating hostel room allocation, fee tracking, and student records management for college administrators.
- **Reduced manual administrative workload by 40%** through streamlined data entry, automated reporting, and role-based access control.

EDUCATION

Integrated M.Sc. in Computer Science — Artificial Intelligence & Machine Learning *Expected Jun 2026*
Nehru Arts and Science College, Kanhangad, Kerala GPA: 7.2 / 10

Relevant Coursework: Machine Learning · Deep Learning · Data Structures & Algorithms · Database Management Systems · Web Technologies

Higher Secondary Education (Class XII — Science) *Mar 2021*
Government Higher Secondary School, Vellur Percentage: 78.67%

Secondary Education (SSLC — Class X) *Mar 2019*
Government Higher Secondary School, Vellur Percentage: 93.3%

CERTIFICATIONS

DeepLearning.AI TensorFlow Developer Certificate — Coursera
Certificate Link *Jun 2025*

Python for Everybody (University of Michigan) — Coursera
Certificate Link *Nov 2021*

CORE STRENGTHS

Strong analytical and problem-solving mindset ·
Effective communicator and collaborative team member ·
Detail-oriented and self-directed learner ·
Adapts quickly to new tools, frameworks, and environments